

Generalizing the Phase-Diversity Algorithm to Nonlinear Excitation

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1 Derivation of the Gradient

The goal will be to understand the effects of nonlinear excitation:

$$s[\phi] = |\mathcal{F}(H)|^{2n} = |\mathcal{F}(pe^{i\phi})|^{2n}$$

where $n \in \mathbb{N}$ is 1 for one-photon excitation, 2, for two-photon excitation, etc. We begin by writing the cost functional, with the minimizer $\hat{F}(\mathbf{u})$ already plugged in:

$$J[\phi] = -\frac{1}{2} \iint_{\mathbb{R}^2} \left[\frac{|\sum_{k=1}^K D_k^* S_k[\phi]|^2}{\gamma + \sum_{k=1}^K |S_k[\phi]|^2} \right] d\mathbf{u} + \text{Constant} + \text{Regularization} \quad (1)$$

We define, for convenience:

$$P[\phi] = \sum_{k=1}^K D_k^* S_k[\phi] \quad (2)$$

$$Q[\phi] = \gamma + \sum_{k=1}^K S_k^*[\phi] S_k[\phi] \quad (3)$$

Which allows us to rewrite the cost functional in [Equation 1](#) as follows:

$$J[\phi] = -\frac{1}{2} \iint_{\mathbb{R}^2} \frac{P[\phi] P^*[\phi]}{Q[\phi]} \quad (4)$$

We will attempt to apply the Gateaux derivative in the direction ψ :

$$\delta_\psi J[\phi] = \lim_{\tau \rightarrow 0} \frac{J[\phi + \tau\psi] - J[\phi]}{\tau} = \frac{d}{d\tau} J[\phi + \tau\psi] \Big|_{\tau=0} \quad (5)$$

We start with:

$$\delta_\psi J[\phi] = \delta_\psi \left(-\frac{1}{2} \iint_{\mathbb{R}^2} \frac{P[\phi] P^*[\phi]}{Q[\phi]} \right) \quad (6)$$

Using the quotient rule, we get:

$$\begin{aligned} d_\psi \iint \frac{P[\phi] P^*[\phi]}{Q[\phi]} &= \iint \frac{Q[\phi] \delta_\psi (P[\phi] P^*[\phi]) - P[\phi] P^*[\phi] \delta_\psi (Q[\phi])}{Q[\phi]^2} \\ &= \iint \frac{Q \left[P^* \sum_{k=1}^K D_k^* \delta_\psi S_k + P \sum_{k=1}^K D_k \delta_\psi S_k^* \right] - P P^* \sum_{k=1}^K (S_k^* \delta_\psi S_k + S_k \delta_\psi S_k^*)}{Q^2} \\ &= \sum_{k=1}^K \iint \frac{P^* Q D_k^* - P P^* S_k^*}{Q^2} \delta_\psi S_k + \text{c.c} \\ &= \sum_{k=1}^K \langle \delta_\psi S_k[\phi], V_k \rangle + \text{c.c} \end{aligned} \quad (7)$$

Where c.c denotes the complex conjugate of the preceeding term, and V_k is define as:

$$V_k = \frac{PQ^*D_k - |P|^2S_k}{Q^2} = F^*D_k - |F|^2S_k \quad (8)$$

Now we expand each of these terms, beginning with Parseval's theorem:

$$\begin{aligned} \langle \delta_\psi S_k[\phi], V_k \rangle &= \langle \delta_\psi s_k, v_k \rangle \\ &= \langle \delta_\psi |h_k|^{2n}, v_k \rangle \\ &= \langle n|h_k|^{2n-2}(h_k^*\delta_\psi h_k + h_k\delta_\psi h_k^*), v_k \rangle \end{aligned} \quad (9)$$

Adding the complex conjugate, we get that each term of the summation in [Equation 7](#) is:

$$\begin{aligned} \langle \delta_\psi S_k, V_k \rangle + \langle V_k, \delta_\psi S_k \rangle &= \langle n|h_k|^{2n-2}(h_k^*\delta_\psi h_k + h_k\delta_\psi h_k^*), v_k \rangle + \langle v_k, n|h_k|^{2n-2}(h_k^*\delta_\psi h_k + h_k\delta_\psi h_k^*) \rangle \\ &= n\langle |h_k|^{2n-2}(h_k^*\delta_\psi h_k + h_k\delta_\psi h_k^*), v_k \rangle + n\langle |h_k|^{2n-2}(h_k\delta_\psi h_k^* + h_k^*\delta_\psi h_k), v_k^* \rangle \\ &= n\langle |h_k|^{2n-2}(h_k^*\delta_\psi h_k + h_k\delta_\psi h_k^*), 2\Re(v_k) \rangle \\ &= 2n\langle 2\Re(h_k^*\delta_\psi h_k), |h_k|^{2n-2}\Re(v_k) \rangle \\ &= 4n\Re(\langle h_k^*\delta_\psi h_k, |h_k|^{2n-2}\Re(v_k) \rangle) \\ &= 4n\Re(\langle \delta_\psi h_k, |h_k|^{2n-2}h_k\Re(v_k) \rangle) \\ &= 4n\Re(\langle \delta_\psi h_k, r_k \rangle) \end{aligned} \quad (10)$$

Where r_k is defined as

$$r_k = |h_k|^{2n-2}h_k\Re(v_k) \quad (11)$$

Now, we note that $h_k = \mathcal{F}^{-1}(H_k)$, where $\delta_\psi H_k = \delta_\psi(pe^{i(\phi+\theta_k)}) = iH_k\psi$. This gives:

$$\langle \delta_\psi S_k, V_k \rangle + \langle V_k, \delta_\psi S_k \rangle = 4n\Re(\langle \mathcal{F}^{-1}(iH_k\psi), r_k \rangle) \quad (12)$$

Applying Parseval's again, we get:

$$\langle \delta_\psi S_k, V_k \rangle + \langle V_k, \delta_\psi S_k \rangle = 4n\Re(\langle iH_k\psi, \mathcal{F}(r_k) \rangle) \quad (13)$$

$$\begin{aligned} &= 4n\Re(\langle iH_k\psi, \mathcal{F}(r_k) \rangle) \\ &= -4n\Im(\langle H_k\psi, \mathcal{F}(r_k) \rangle) \\ &= -4n\Im(\langle \psi, H_k^*\mathcal{F}(r_k) \rangle) \\ &= 4n\langle \psi, \Im(H_k^*\mathcal{F}(r_k)) \rangle \end{aligned} \quad (14)$$

where the last move is allowed since ψ is completely real. Plugging back in to [Equation 6](#), we get:

$$\begin{aligned} \delta_\psi J[\phi] &= -\frac{1}{2} \sum_{k=1}^K 4n\langle \psi, \Im(H_k^*\mathcal{F}(r_k)) \rangle \\ &= \langle \psi, -2n \sum_{k=1}^K \Im(H_k^*\mathcal{F}(r_k)) \rangle \end{aligned} \quad (15)$$

Now, we note the definition of the gradient:

$$\delta_\psi J[\phi] = \langle \psi, \nabla J[\phi] \rangle \quad (16)$$

Inspecting [Equation 15](#), we conclude:

$$\nabla J[\phi] = -2n \sum_{k=1}^K \Im(H_k^*\mathcal{F}(r_k)) \quad (17)$$

$$= -2n \sum_{k=1}^K \Im(H_k^*\mathcal{F}(|h_k|^{2n-2}h_k\Re(\mathcal{F}^{-1}(V_k)))) \quad (18)$$

From here on, we will refer to this gradient as $g[\phi]$.

2 Derivation of the Hessian

We will begin with the assumption that the phase ϕ can be expanded over a basis $\{Z_i\}$:

$$\phi(\mathbf{x}) = \sum_i \langle \phi, Z_i \rangle Z_i(x) \quad (19)$$

The goal now is to find the discrete Hessian. If there are m basis functions in our expansion, we write its i, j th entry as:

$$\mathbf{H}_{ij} = \langle H[\phi]Z_j, Z_i \rangle, \quad 1 \leq i, j \leq m \quad (20)$$

Where the Hessian entries can be approximated by the finite differences of the gradient:

$$H[\phi]Z \approx \frac{g[\phi + \tau Z] - g[\phi]}{\tau}, \quad \tau \text{ small} \quad (21)$$

Now, we will attempt to find the Gauss-Newton approximation of $H[\phi]Z$ (the action of the Hessian on the basis element Z). Begin by rewriting the cost functional in [Equation 1](#) as follows:

$$J[\phi] = \frac{1}{2} \sum_{k=1}^K \sum_{j=1}^{k-1} \|R_{jk}[\phi]\|^2 + \frac{\gamma}{2} \|D^{1/2}[\phi]\|^2 \quad (22)$$

Where we define:

$$R_{jk}[\phi] = \frac{D_k S_j[\phi] - D_j S_k[\phi]}{Q^{1/2}[\phi]} \quad (23)$$

$$D[\phi] = \frac{\sum_{k=1}^K |D_k|^2}{Q[\phi]} \quad (24)$$

$$Q[\phi] = \gamma + \sum_{k=1}^K S_k^*[\phi] S_k[\phi] \quad (25)$$

We begin by dropping the term on the RHS of [Equation 22](#). This gives us a simple least-squares cost function, which allows us to write the Gauss-Newton approximation as follows:

$$\langle H_{GN}[\phi]Z, \psi \rangle \approx \sum_{k=1}^K \sum_{j=1}^{k-1} \langle R'_{jk}[\phi]Z, R'_{jk}[\phi]\psi \rangle \quad (26)$$

It is helpful now to redefine some variables:

$$\tilde{D}_k = \frac{D_k}{Q^{1/2}} \quad (27)$$

$$\tilde{R}_{jk}[\phi] = \tilde{D}_k S_j[\phi] - \tilde{D}_j S_k[\phi] \quad (28)$$

$$U_{jk} = \tilde{R}'_{jk}[\phi] = \tilde{D}_k S'_j[\phi]Z - \tilde{D}_j S'_k[\phi]Z \quad (29)$$

As we know from the gradient calculation, Q is not fixed with respect to ϕ . However, we make this approximation in order to ease the rest of the derivation. Using our new definitions, we write:

$$\langle H_{GN}[\phi]Z, \psi \rangle = \sum_{k=1}^K \sum_{j=1}^{k-1} \langle U_{jk}, \tilde{R}'_{jk}[\phi]\psi \rangle \quad (30)$$

Now we approximate $S'_j[\phi]Z$ for fixed $Z \in L^2(\mathbb{R}^2)$. We drop the subscript j and let W be arbitrary. Then, we get:

$$\begin{aligned} \langle S'[\phi]Z, W \rangle &= \langle \delta_Z S[\phi], W \rangle \\ &= \langle \delta_Z s[\phi], w \rangle \\ &= \langle \delta_Z |h|^{2n}, w \rangle \\ &= \langle n|h|^{2n-2}(h^* \delta_Z h + h \delta_Z h^*), w \rangle \end{aligned} \quad (31)$$

Now, we again note that $h = \mathcal{F}^{-1}(H)$ where $\delta_Z H = iHZ$, which gives $\delta_Z h = \mathcal{F}^{-1}(iHZ)$ This gives:

$$\begin{aligned}\langle S'[\phi]Z, W \rangle &= n\langle i|h|^{2n-2}h^*\mathcal{F}^{-1}(HZ), w \rangle + n\langle -i|h|^{2n-2}h(\mathcal{F}^{-1}(HZ))^*, w \rangle \\ &= \langle -2n\Im(|h|^{2n-2}h^*\mathcal{F}^{-1}(HZ)), w \rangle\end{aligned}\quad (32)$$

Which allows us to identify:

$$S'[\phi]Z = -2n\mathcal{F}(\Im(|h|^{2n-2}h^*\mathcal{F}^{-1}(HZ))) \quad (33)$$

Giving:

$$U_{jk} = -2n\tilde{D}_k\mathcal{F}(\Im(|h_j|^{2n-2}h_j^*\mathcal{F}^{-1}(H_jZ))) + 2n\tilde{D}_j\mathcal{F}(\Im(|h_k|^{2n-2}h_k^*\mathcal{F}^{-1}(H_kZ))) \quad (34)$$

Now we define:

$$W_{jk,i} = \tilde{D}_i^*U_{jk} \quad (35)$$

$$w_{jk,i} = \mathcal{F}^{-1}(\tilde{D}_i^*U_{jk}) \quad (36)$$

$$(37)$$

Now, examining [Equation 30](#), we unpack each of the terms.

$$\langle U_{jk}, \tilde{D}_k S'_j[\phi]\psi \rangle = \langle S'_j[\phi]^* W_{jk,k}, \psi \rangle \quad (38)$$

$$= \langle 2n\Im(H_j^*\mathcal{F}(|h_j|^{2n-2}h_j\mathcal{F}^{-1}(W_{jk,k}))) , \psi \rangle \quad (39)$$

Looking back at [Equation 30](#), we can identify:

$$H_{GN}[\phi]Z = 2n \sum_{k=1}^K \sum_{j=1}^{k-1} \Im \left[H_j^* \mathcal{F} \left(|h_j|^{2n-2} h_j \mathcal{F}^{-1}(\tilde{D}_k^* U_{jk}) \right) \right] - \Im \left[H_k^* \mathcal{F} \left(|h_k|^{2n-2} h_k \mathcal{F}^{-1}(\tilde{D}_j^* U_{jk}) \right) \right] \quad (40)$$

To clean this up and match the convention from the original paper, we redefine:

$$\tilde{U}_{jk} = \tilde{D}_j\mathcal{F}(\Im(|h_k|^{2n-2}h_k^*\mathcal{F}^{-1}(H_kZ))) - \tilde{D}_k\mathcal{F}(\Im(|h_j|^{2n-2}h_j^*\mathcal{F}^{-1}(H_jZ))) \quad (41)$$

Which gives:

$$H_{GN}[\phi]Z = 4n^2 \sum_{k=1}^K \sum_{j=1}^{k-1} \Im \left[H_j^* \mathcal{F} \left(|h_j|^{2n-2} h_j \mathcal{F}^{-1}(\tilde{D}_k^* \tilde{U}_{jk}) \right) \right] - \Im \left[H_k^* \mathcal{F} \left(|h_k|^{2n-2} h_k \mathcal{F}^{-1}(\tilde{D}_j^* \tilde{U}_{jk}) \right) \right] \quad (42)$$